F00-SR-001
PLANT 33kV HV DISTRIBUTION SWITCHROOM
FROM SWITCHROOM E-SHUT F00SR001-LCS (NOTE 9)
DRG 2000-F00-SLD-EL-10003

F00-SW-010
PLANT 33kV HV DISTRIBUTION SWITCHBOARD

CONT'D FROM SHEET 4, DRG 2000-F00-SLD-EL-10005

33kV INCOMER 3
FROM PARKESTON POWER STATION
FIM C (POLE FC69)

NOTE 3
2000/1A 2.5VA CL0.2
TEST
PQM SEL 735
2000/1A 2.5VA 5P20
NOTE 1
INTERTRIP TO RECLOSER (REFER NOTE 10)
SWITCHROOM E-SHUT (REFER NOTE 9)
CLOSE
TRIP
PR SEL 751
50 50G 51 51G AFD 3
M 2000A INC-3 2000A

36kV 2000A 3-E 31.5kA FOR 3s

B22 B23 B24 B25 B26

M 630A
300/1A 2.5VA 5P20
300/1A 1VA 10P10
NOTE 2

VT
33kV/√3 0.1kV/√3 0.1kV/3
35VA CLASS 0.2
35VA CLASS 3P

ACTIVATION OF SWITCHROOM E-SHUT (REFER NOTE 4)
INTERTRIP FROM 11kV SWITCHBOARD INCOMER (REFER NOTE 8)
ACTIVATION OF SWITCHROOM E-SHUT (REFER NOTE 4)
DETECTION OF EARTH FAULT (REFER NOTE 5)
DETECTION OF ARC FAULT (REFER NOTE 6)
ACTIVATION OF SWITCHROOM E-SHUT (REFER NOTE 4)
DETECTION OF EARTH FAULT (REFER NOTE 5)
DETECTION OF ARC FAULT (REFER NOTE 6)

F23-TR-016
FINAL TAILS PUMPS
33/11kV TRANSFORMER

F23-SW-016
FINAL TAILS PUMPS
11kV SWITCHBOARD INCOMER

F23-TR-023
FINAL TAILS PUMPS
33/0.433kV TRANSFORMER

F23-TR-023
FINAL TAILS PUMPS
415V MCC INCOMER

F13-TR-020
MILL CHILLERS & REAGENTS
33/0.433kV TRANSFORMER

F13-SW-020
MILL CHILLERS & REAGENTS
415V MCC INCOMER

REFER SLD DRG 2000-F23-SLD-EL-10001
REFER SLD DRG 2000-F23-SLD-EL-10201
REFER SLD DRG 2000-F13-SLD-EL-10701

33kV SPARE (3)
FULLY EQUIPPED
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NOTES

1. THE FEEDER PROTECTION RELAY VOLTAGE INPUT SHALL BE WIRED FROM THE CORRESPONDING INCOMER BUS VT.
2. ARC FLASH POINT DETECTORS SHALL BE PLACED IN CABLE ZONES AND CONNECTED DIRECTLY TO THE SEL PROTECTION RELAYS IN EACH PANEL.
3. SWITCHBOARD SHALL BE EXTENSIBLE FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER PER BUS SECTION AS SHOWN.
4. TRANSFORMER 33KV FEEDER SHALL BE TRIPPED ON THE ACTIVATION OF THE SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP THE CIRCUIT BREAKER.
5. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE EARTH FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
6. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ARC FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
7. TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS SHALL BE HARDWIRED TO THE TRANSFORMER FEEDER PROTECTION RELAY.
8. THE INTERTRIP OF THE UPSTREAM 33kV FEEDER SHALL BE HARDWIRED FROM THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY EITHER ON A TRANSFORMER TRIP, EARTH FAULT DETECTION OR ARC FAULT DETECTION.
9. THE ACTIVATION OF THE PLANT 33KV DISTRIBUTION SWITCHROOM F00-SR-001 EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP ALL THREE INCOMER CIRCUIT BREAKERS AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSERS.
10. PROTECTION TRIP FROM THE INCOMER CIRCUIT BREAKER SHALL INTERTRIP THE UPSTREAM POLE MOUNTED RECLOSER.

LEGEND

- PR — INCOMER / FEEDER PROTECTION RELAY NOTES 1 & 2
- PQM — POWER QUALITY AND REVENUE METERING
- ARC FLASH SENSOR (POINT TYPE) NOTE 2
- CAPACITIVE VOLTAGE INDICATION
- MECHANICAL OR ELECTRICAL INTERLOCK DEVICE
- 27 — UNDER-VOLTAGE
- 50 — INSTANTANEOUS OVERCURRENT
- 50G 50N — INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
- 51 — TIMED OVERCURRENT
- 51G 51N — TIMED EARTH/NEUTRAL OVERCURRENT
- 64 — EARTH FAULT
- 67 67G — DIRECTIONAL OVERCURRENT/EARTH FAULT
- AFD — ARC FLASH DETECTION

NORTHERN STAR

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F00-SLD-EL-10006_Rev3

Single line diagram sheet 5